

## Elektor AXL V2 with Lateral MOSFET's

 **HerrFlick** ·  2019-01-14 3:09 pm



**HerrFlick**

Member

Joined 2017

2019-01-14 3:09 pm

  #1

More then 30 years ago (1985) I build the Elektor AXL class A/AB amplifier, time to upgrade this amplifier using new (= better) components.

My main goal was to replace the power mosfets 2SK134/2SJ49 with Exicon ECF10N20/ECF10P20. Also the medium power transistor BF469/BF470 changed to MJE340G/MJE350G. Extra power supply filter capacitors where used and placed on the PCB.

The two capacitors for the 15V zener diodes where increased to 100uF 25V

and the capacitor parallel to the quiescent current potentiometer (1k) is increased to 220  $\mu$ F (Single-Ended Conductive Polymer Aluminum Solid Electrolytic Capacitors) plus extra 47nF WIMA MKP. The two source resistors (0,22 Ohm) are no longer needed.

The input capacitor (C1,C2,C3-820 nF) is replaced by a high end capacitor 2,7 uF Clarity Cap ESA.

The Exicon Application notes told me to decrease the inductance in the output line to 1 $\mu$ H (1mm Cu wire inner diameter 8 mm, 15 windings). Due to different gate input capacitance of the mosfets the gate resistors are also different, 330 Ohm for ECF10N20 and 220 Ohm for ECF10P20.

The power supply for each channel (2x25V AC 2,4A 120 VA) is filtered by 4 x 10.000  $\mu$ F rectifier (Ebay) so that results in 2 x 38V DC output voltage.

It is still very important to set the quiescent current potentiometer to zero Ohm before switching on the amplifier for the first time.

I tried to use the same component labels but did not 100 percent succeed.

On my old (same as me) but still working oscilloscope (Philips

PM3226) I saw a maximum error free waveform of about 60 Volt peak-peak with a 8 Ohm resistor load.  
No oscillation what so ever, 60 Volt peak to peak over 8 Ohm i.e.  $30 : 1,4 = 22$  Watt output power with a quiescent current 0,1 A.  
The calculated damping factor is more then 200.  
I am using a Rasberry Pi in combination with the HiFiBerry DAC plus and a home made power supply (5V 3A), controlled by Volumio software, NAS. The loudspeakers: Acoustic Energy AE1.

### Attachments

[\(URL: /community/attachments/axl-v2-schema-jpg.728956/\)](URL: /community/attachments/axl-v2-schema-jpg.728956/)

AXL-v2-schema.jpg

303.4 KB · Views: 1,645

[\(URL: /community/attachments/axl-v2-assambled-jpg.728957/\)](URL: /community/attachments/axl-v2-assambled-jpg.728957/)

AXL-v2-assambled.jpg

701 KB · Views: 1,703

[\(URL: /community/attachments/axl-v2-pcb-scl-jpg.728961/\)](URL: /community/attachments/axl-v2-pcb-scl-jpg.728961/)

AXL-v2-PCB-scl.jpg

588.1 KB · Views: 1,473

[\(URL: /community/attachments/axl-v2-pcb-bcl-jpg.728962/\)](URL: /community/attachments/axl-v2-pcb-bcl-jpg.728962/)

AXL-v2-PCB-bcl.jpg

536.3 KB · Views: 1,260

[\(URL: /community/attachments/axl-v2-jpg.728963/\)](URL: /community/attachments/axl-v2-jpg.728963/)

AXL-v2.jpg

707.9 KB · Views: 1,366

Last edited: 2019-02-19 12:58 pm



**mlloyd1**

Member

Joined 2001

2019-01-14 4:49 pm

  #2

are you sure there are no errors in that schematic?  
some things around the drive for t11 and t12 looks a bit strange ...

mlloyd1

|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>H</div> <div><b>Herr Flick</b></div> <div>Member</div> <div>Joined 2010</div> | <div>2019-01-14 8:00 pm</div> <div>  #3</div> <div>Thanks for your reply -mlloyd1-.</div> <div>I checked again my schema with the original one from march 1985 but I assure you there is no difference at all. I also check the pcb layout again and came to the same conclusion. If there is a problem then my amplifier could probably not produce such a clear sound at all I think.</div> <div>Herr Flick</div> |
| <div>nigelwright7557</div> <div>Member</div> <div>Joined 2008</div>                | <div>2019-02-28 12:37 am</div> <div>  #4</div> <div>Completely over engineered for very little improvement, if any.</div> <div>Newer doesn't always mean better, could be just "different" or even "the same"</div>                                                                                                                                                                                                 |
| <div>J</div> <div><b>johnego</b></div> <div>Member</div> <div>Joined 2017</div>    | <div>2019-02-28 1:36 am</div> <div>  #5</div> <div>Herr Flick,</div> <div>How would you compare the sound quality of this new version with the old one? And with other amplifiers?</div>                                                                                                                                                                                                                          |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="248 174 296 232">H</div> <div data-bbox="205 282 341 313"><b>HerrFlick</b></div> <div data-bbox="173 333 274 356">Member</div> <div data-bbox="159 369 288 392">Joined 2017</div>             | <div data-bbox="461 152 756 183">2019-02-28 3:30 pm</div> <div data-bbox="1295 152 1453 183">  #6</div> <div data-bbox="461 226 1445 568"><p>It is always difficult to find the right words to describe the difference between two amplifiers -jonnego-, especially when you are not a native english speaker. The first thing we (my wife and I) said to each other when we listen to my upgraded amplifier was: much more details and a lower bass. A couple of weeks ago I went to a friend and made comparision with his amplifier NAD C352, Bowers and Wilkens 804S loudspeakers and -van den Hul D-101 III interlinks.</p></div> <div data-bbox="461 584 1431 840"><p>All three of us came to the conclusion that the Elektor AXL V2 was slightly "better" i.e. more details. Later on I invited another friend, HiFi freak as well, at home and he was impressed by the sound. Another big improvement was earlier made due to the use of the Raspberry Pi and the HiFiBerry DAC, thatswhy I sold my Arcam Alpha 7 CD player.</p></div> <div data-bbox="461 853 1445 974"><p>I will try to make more comparings with other Hi-End amplifiers and let you know by that time, but it is hard to find people who uses Hi-End stereo equipment nowadays.</p></div> |
| <div data-bbox="253 1108 288 1167">J</div> <div data-bbox="213 1216 331 1247"><b>johnego</b></div> <div data-bbox="173 1267 274 1290">Member</div> <div data-bbox="159 1303 288 1326">Joined 2017</div>       | <div data-bbox="461 1088 745 1120">2019-03-01 6:51 am</div> <div data-bbox="1295 1088 1453 1120">  #7</div> <div data-bbox="496 1182 1339 1218"><b>HerrFlick said:</b> (<i>URL: /community/goto/post?id=5713404</i>)</div> <div data-bbox="496 1247 1310 1321"><p>it is hard to find people who uses Hi-End stereo equipment nowadays.</p></div> <div data-bbox="461 1413 1426 1543"><p>thanks herr flick. seems like you are new with diy. nowadays, if we are selective we can build amp that rivals commercial amps many times the price of the NAD</p></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <div data-bbox="248 1677 296 1736">S</div> <div data-bbox="165 1785 381 1816"><b>sariwangi2008</b></div> <div data-bbox="221 1836 322 1859">Member</div> <div data-bbox="207 1872 336 1895">Joined 2016</div> | <div data-bbox="461 1657 751 1688">2019-03-04 3:19 am</div> <div data-bbox="1295 1657 1453 1688">  #8</div> <div data-bbox="461 1729 1015 1850"><p>What amp do you recommended?<br/>With latfet 2SK176/2SJ56.<br/>Its worthed if I build AXL mosfet now?</p></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

H

**HerrFlick**Member  
Joined 2017

2019-03-04 3:05 pm

  #9

The 2SK176/2SJ56 is a good replacement for the original 2SK134/2SJ49, but you have to use the old Elektor schema (1985). I.e. you have to insert two source resistors (0,22 Ohm low inductance) and lower the gate resistor R25 330 Ohm in 220 Ohm. The output coil L1 must be changed back in 2  $\mu$ H. Because of the source resistors you need a new PCB layout and last but not least, I think the the sound of the lateral mosfets ECF10N20/ECF10P20 is better but I cannot confirm that.

P

**phri**Member  
Joined 2019

2019-03-25 3:04 am

  #10**upgrade AXL****HerrFlick said:** (URL: /community/goto/post?id=5718064)

The 2SK176/2SJ56 is a good replacement for the original 2SK134/2SJ49, but you have to use the old Elektor schema (1985). I.e. you have to insert two source resistors (0,22 Ohm low inductance) and lower the gate resistor R25 330 Ohm in 220 Ohm. The output coil L1 must be changed back in 2  $\mu$ H. Because of the source resistors you need a new PCB layout and last but not least, I think the the sound of the lateral mosfets ECF10N20/ECF10P20 is better but I cannot confirm that.

What was the motivation to replace the mosfets? In your opinion which upgrade improved sound quality most?

I build 8 versions of the AXL back in the 80's, and still have them, stored in the shed (next to fancy also unused electrostatic speakers), as they have been replaced with a modern Yamaha AV-receiver. However I have some time for the Audio hobby again and look at options to use these again.

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Kay Pirinha</b><br/>Member<br/>Joined 2011</p> | <p>2019-03-28 12:33 pm</p> <p>🔗 📌 #11</p>                                                                                                                                                                                                                                                                                                                                               |
|                                                      | <div data-bbox="464 224 1453 412"><p><b><u>HerrFlick said:</u></b> (<i>URL: /community/goto/post?id=5664508</i>)</p><p>60 Volt peak to peak over 8 Ohm i.e. <math>30 : 1,4 = 22</math> Watt output power with a quiescent current 0,1 A.</p></div> <p>No. It calculates to <math>60 \text{ Vpp}^2 / (8 \times 8 \Omega) = 56.25 \text{ W}</math>.</p> <p>Congrats and best regards!</p> |



**HerrFlick**

Member

Joined 2017

2019-05-12 10:11 am

↩ 📌 #12

Hi Kay, sorry for my late answer.

I build this amplifier more than 35 years ago, and as you know a lot of components will deteriorate during time, especially the electrolytic capacitors and nowadays one can buy much better replacements.

That's why I started to upgrade my old Elektor AXL amplifier.

But before that I only replaced the input capacitors 3x 820nF MKM with a 2,7µF MKP Clarity Cap.

That was a good improvement, much more details. I also wanted to use a better power supply capacitor with low ESR near the power transistor.

So I had to redesign the PCB layout. I read somewhere in this - diyaudio- forum that a good replacement for the BF470/BF469 is MJE340/MJE350.

The power mosfets 2SK134/2SJ49 where replaced by Exicon ECF10N20/ECF10P20 because they have a low distortion, very fast slew rates

and free from secondary breakdown and thermal runaway.

For a long-time stable capacitors I replaced C5/C6/C9 with Single-Ended Conductive Polymer Aluminum Solid Electrolytic Capacitors. Very critical capacitors are the one in the snubber network C18, the best is a FKP3 (Wima), and last but not least, the power supply capacitor where increased to 20.000 µF per channel.

I must say it was really worth to do this investment, I invited some HiFi freaks and the where all impressed by the excellent sound quality of

and this elektor AXL V2 amplifier. I am using a Raspberry Pi with a HiFiBerry DAC on top, directly connected to this amplifier.

More info:

Exicon Application Notes

**<https://www.wima.de/wp-content/uploads/media/WIMA-Audio.pdf>** (URL: <https://www.wima.de/wp-content/uploads/media/WIMA-Audio.pdf>)

**[HiFiBerry | Audiophile 24/96, 24/192 sound cards \(Audio DAC and SPDIF out\) for the Raspberry Pi, audio out, audio cards, audio projects](https://www.hifiberry.com/)** (URL: <https://www.hifiberry.com/>)

p.s. you're right, 56 Watt RMS power, enough for an average living room.

**HerrFlick**

Member

Joined 2017

2019-05-13 1:01 pm

  #13

I forgot another important internet site, it is all about highend capacitors. A very extensive capacitor test with a lot of information:

**Humble Homemade Hifi - Cap Test (URL:**  
**<http://www.humblehomemadehifi.com/Cap.html>**

Best regards!

P

phri

Member

Joined 2019

2019-05-18 4:19 am

  #14**HerrFlick said:** (URL: </community/goto/post?id=5788635>)

Hi Kay, sorry for my late answer.

I build this amplifier more than 35 years ago, and as you know a lot of components will deteriorate during time, especially the electrolytic capacitors and nowadays one can buy much better replacements.

That's why I started to upgrade my old Elektor AXL amplifier.

But before that I only replaced the input capacitors 3x 820nF MKM with a 2,7µF MKP Clarity Cap.

My upgrades which I did many years ago where a bit different.

>After making the circuit boards I put a layer of silver on top of the copper, to reduce risk of corrosion.

>I bypassed the input capacitors C1, 2, 3) completely; thus there is no capacitor directly in the signal pathway. btw I did the same with the output of my pre-amp, thus only cap is at the input pre-amp

> I separated the power supply of the 2 powerfets from the rest. I lifted the power supply with +/-5V (to +/- 45V) for the part which takes care of the voltage amplification, with it's own rectifier and Elco's. While the 2 powerfets were kept at +/- 40 V. Thus increased the headroom of the amp, while keeping the dissipation relative low. This also improves signal to noise ratio. The main reason I did this was because the units were driving an electrostatic speaker with a relative low efficiency. Bias setting is around 0.6 A

> I replaced each of the series resistors for the power mosfets with 5 non inductive film resistors.

My AXL's still play ok with all the old capacitors. Although I do worry a bit about over-voltage as the units were build for dutch 220 V AC, while we often have 250 - 260 V AC here in aussieland mainly due to the popularity of solar panels which push up grid voltage.

H

**HerrFlick**

Member

Joined 2017

2019-05-18 10:24 am

  #15

Well -phri- that's one of the reason I reconstructed my amplifier. My old PCB was homemade, drew with a pencil and was only a copper surface and the soldering tin was old fashioned with lead. My new designed PCB was professional made and I used tin/silver soldering. To bypass the input capacitor is a tricky thing but as long as you use your amplifier with a well known sound source then you have a advantage because the input capacitor got a great influence on the overall sound quality. That's why I used a high end capacitor instead. Yes, to drive a electrostatic loudspeaker you need a lot of power. But the sound quality is excellent but often a little lack of bass. I would like to tune my elektor AXL V2 for instance by raising the quiescent current to 200mA for less distortion. And I have to determine the gate resistor values as well. One method to determine the right gate resistor value is to overdrive the amplifier with a square wave. The shape of the rise and fall curves will indicate a correct value. A very good application note with lot of information is: <http://www.ti.com/lit/an/snaa045a/snaa045a.pdf> (URL: <http://www.ti.com/lit/an/snaa045a/snaa045a.pdf>)

But first holidays!

In the Netherlands by the way is nowadays the main voltage 230 V 50 Hz.

P

phri

Member

Joined 2019

2019-05-21 5:32 am

  #16

I had a look at the paper on mosfets mentioned in the earlier message, good stuff, although the old mosfets in the original AXL design are not listed; 2SK135 & 2SJ50

I started measuring the setting on my old build. The AXL I build as mono-blocks run at 93 V now and a bias of around 0.6 A, thus a bit over 25 W dissipation for each fet. Elektor stated a max total dissipation of 65W. Datasheets of the mosfets state 100 W each thus I should be fine. The datasheet also mentioned a drain-source voltage of max 160V for the 2SK135 thus below 100V shouldn't be a problem.

btw, the units do get really hot after running for an hour or so; scientifically..... touching a mosfet longer then 2 seconds is uncomfy.

I also got my first build of the AXL back which is a stereo version build unmodified with the original elekor prints. This unit can oscillate now and then probably due to the wirewound resistors in series with the mosfets. It runs at 80 V with a bias of 200 mA. If I push this unit; sound is less clean compared to the others. maybe due to working in AB class?? I am not sure. This stereo unit is also extreme reliable as it lived at a friends place running daily for more then 20 years. However always had (and still has) a buzzing power supply; toriodal transformer (300 W & 2 X 51000 uF) which really annoys me.

Overall sound quality of this 35 year old amp design (the AXL of-course) build by a hobbyist (me) still outperforms my modern \$2000 yamaha surround sound receiver, and the latter got stellar reviews from the experts...

H

Herr Flick

Member

Joined 2010

2019-05-21 4:45 pm

  #17

Wow, 2 x 51000  $\mu$ F that a lot capacity. Did you use a rubber slice under and on top of the toroidal transformer ? Nice to read about the outperforming Yamaha receiver. My AXL also had a slighty better sound compared to the NAD C 352 stereo amplifier.

**metanastis**

Member

Joined 2009

2020-05-05 3:21 pm

  #18

I constructed my AXL in 1989. Its operation is in class A- 1A current without input because quality matters. I have used it to so many parties as a D.J and it has never denied neither quality or strength. Now it is my main equipment for testing the tube preamplifiers I build. My power supply caps are 40000uF/63volts and supply the amplifier through a darlington stabilization circuit, giving it +- 38volts. This amplifier just stays a little bit before my Audio Note class A tube power amplifier and I use it in every party.

**HerrFlick**

Member

Joined 2017

2020-05-23 7:18 pm

#19

**On/Off switch LED pwr supply**

Nice to hear from you -Metanastics- What a pity we live so far from each other. I would like to make a direct compare between your old AXL amplifier and my upgraded AXL V2 amplifier. I recently made a tiny modification for the ON/OFF switch LED illumination. It was powered by a old fashioned small powersupply with a 1,8VA transformer and 100mA stablized 12V out. But I realized that it would be more usefull to have a more "intelligence" one. This signal LED in the ON/OFF switch should go ON when all the four power supply voltages (-38V and + 38V) for the left and right channel are in good health. If one of them is down the LED signal should be off. So I used a opto-coupler powered by the four amplifier power supply's. It is just a small PCB which can easily find a place inside the amplifier.

**Attachments**

[\(URL: /community/attachments/axl-on-off-led-pwr-supply-pcb-ipg.846720/\)](#)

AXL ON-OFF LED PWR supply PCB.jpg

128.3 KB · Views: 565

[\(URL: /community/attachments/axl-on-off-led-pwr-supply-schema-jpg.846719/\)](#)

AXL ON-OFF LED PWR supply schema.jpg

74.2 KB · Views: 541

2020-05-23 7:50 pm

  #20**metanastis**

Member

Joined 2009

**HerrFlick said:** (URL: /community/goto/post?id=6216189)

Nice to hear from you -Metanastics- What a pity we live so far from each other. I would like to make a direct compare between your old AXL amplifier and my upgraded AXL V2 amplifier. I recently made a tiny modification for the ON/OFF switch LED illumination. It was powered by a old fashioned small powersupply with a 1,8VA transformer and 100mA stablized 12V out. But I realized that it would be more usefull to have a more

I can assure you that my old AXL is playing superbly so far!!! Of course i haven't listened to yours so I can't compare each other, but, my power supply is not the one given by Elektor. There is a pcb stabilization circuit in darlington connection which kills any ac voltage if there were any!!! Pure dc means clear sound plus the 40000uF capacitors = excellence!! Now i am trying to make a new circuit board using sprint layout 6.0 'cause the old pcbs were been made in an amateur way and they need renovation!! As far as the circuit you have made, it is a very good idea for protecting the speakers. Audio Research and a couple of other companies used to place optocouplers at their tube preamplifiers in late 80's- early 90's. Good job.!!! Personally I don't deal with transistors, just tubes! I have made exact pcb clones of some legends of tube preamplifiers brands as Audio Note L2-M2, Marantz 7C, Audio Research SP8E MKII etc.

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|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>metanastis</b></p> <p>Member</p> <p>Joined 2009</p>                 | <p>2020-10-05 4:35 pm <span>🔗 📌 #21</span></p> <div data-bbox="464 224 1453 651"><p><b>HerrFlick said:</b> (URL: /community/goto/post?id=6216189)</p><p>Nice to hear from you -Metanastics- What a pity we live so far from each other. I would like to make a direct compare between your old AXL amplifier and my upgraded AXL V2 amplifier. I recently made a tiny modification for the ON/OFF switch LED illumination. It was powered by a old fashioned small powersupply with a 1,8VA transformer and 100mA stablized 12V out. But I realized that it would be more usefull to have a more</p></div> <p>What's your power transformer specifications in secondary volts and current?</p>                                       |
| <p><b>H</b></p> <p><b>Herr Flick</b></p> <p>Member</p> <p>Joined 2010</p> | <p>2020-10-05 8:28 pm <span>🔗 📌 #22</span></p> <p>It's a Amplimo toroidal transformer 2 x 25V 2,4 A = 120VA</p> <p><b><u><a href="https://www.amplimo.nl/images/downloads/ds_standardrange/48016.pdf">https://www.amplimo.nl/images/downloads/ds_standardrange/48016.pdf</a></u></b> (URL: <u><a href="https://www.amplimo.nl/images/downloads/ds%20standardrange/48016.pdf">https://www.amplimo.nl/images/downloads/ds%20standardrange/48016.pdf</a></u>)</p>                                                                                                                                                                                                                                                                       |
| <p><b>metanastis</b></p> <p>Member</p> <p>Joined 2009</p>                 | <p>2020-10-07 10:39 am <span>🔗 📌 #23</span></p> <p>Try the class A operation- 1A current wit no load. You will be amazed by the sonic impression! I work it to its limits with +- 39VDC 5A power supply. My capacitors are 40000Uf/63VDC and i have built 2 separate boxes for power supply and amplifier. Yesterday I finished the 3 pcbs, 2 for the amp and 1 for the DC stabilization. I have ordered my spare parts from mouser and I will start rebuilding the project under the knowledge and experience of today. 15 years ago I came to your country working for a military project of my company( Hellenic Aerospace Industry).I stayed for 1 month in Winterswijk and I had a really good time with the local fellows.</p> |

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| <b>metanastis</b><br>Member<br>Joined 2009 | <div>2020-10-07 11:47 am <span>🔗 📌 #24</span></div> <p>I would appreciate it If you could inform me about the exact dimensions of the old AXL pcb because scanning the image from the magazine is a little bit tricky.</p> <p>Thanks a lot</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>metanastis</b><br>Member<br>Joined 2009 | <div>2020-10-08 8:19 am <span>🔗 📌 #25</span></div> <div><b>Herr Flick said:</b> (URL: /community/goto/post?id=6365191)<br/>It's a Amplimo toroidal transformer 2 x 25V 2,4 A = 120VA<br/><a href="https://www.amplimo.nl/images/downloads/ds_standardrange/48016.pdf">https://www.amplimo.nl/images/downloads/ds_standardrange/48016.pdf</a> (URL: <a href="https://www.amplimo.nl/images/downloads/ds%20standardrange/48016.pdf">https://www.amplimo.nl/images/downloads/ds%20standardrange/48016.pdf</a>).</div> <p>Try the class A operation- 1A current wit no load. You will be amazed by the sonic impression! I work it to its limits with +- 39VDC 5A power supply. My capacitors are 40000Uf/63VDC and i have built 2 separate boxes for power supply and amplifier. Yesterday I finished the 3 pcbs, 2 for the amp and 1 for the DC stabilization. I have ordered my spare parts from mouser and I will start rebuilding the project under the knowledge and experience of today. 15 years ago I came to your country working for a military project of my company( Hellenic Aerospace Industry).I stayed for 1 month in Winterswijk and I had a really good time with the local fellows.I would appreciate it If you could inform me about the exact dimensions of the old AXL pcb because scanning the image from the magazine is a little bit tricky.</p> <p>Thanks a lot</p> |

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| <p><b>metanastis</b></p> <p>Member</p> <p>Joined 2009</p>           | <p>2020-10-08 11:15 am <span>↩</span> <span>🔖</span> #26</p> <div data-bbox="464 221 1453 651"><p><b><u>HerrFlick said:</u></b> (URL: /community/goto/post?id=6216189)</p><p>Nice to hear from you -Metanastics- What a pity we live so far from each other. I would like to make a direct compare between your old AXL amplifier and my upgraded AXL V2 amplifier. I recently made a tiny modification for the ON/OFF switch LED illumination. It was powered by a old fashioned small powersupply with a 1,8VA transformer and 100mA stablized 12V out. But I realized that it would be more usefull to have a more</p></div> <p>I will give it a try in my new AXL construction-Thanks a lot</p> |
| <p><b>P</b></p> <p><b>phri</b></p> <p>Member</p> <p>Joined 2019</p> | <p>2021-04-10 6:22 am <span>↩</span> <span>🔖</span> #27</p> <div data-bbox="464 889 1453 1160"><p><b><u>metanastis said:</u></b> (URL: /community/goto/post?id=6367042)</p><p>I would appreciate it If you could inform me about the exact dimensions of the old AXL pcb because scanning the image from the magazine is a little bit tricky.</p><p>Thanks a lot</p></div> <p>Did you manage to build the old AXL, if so how does it sounds? Any large deviations from the original design?</p> <p>Currently I use 2 AXL amps daily, connected to a laptop usb to a high quality dac. With Seas Excel homebuild speakers. Sound is truly amazing.</p>                                               |

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| <div data-bbox="188 282 355 313"><b>metanastis</b></div> <div data-bbox="173 331 276 356">Member</div> <div data-bbox="156 367 293 392">Joined 2009</div>                                                      | <div data-bbox="459 152 759 183">2021-04-10 11:26 am</div> <div data-bbox="1278 152 1453 183">  #28</div> <div data-bbox="459 226 1453 1019"><p>I have finally opened the old AXL case and measured the pcb dimensions. I have designed the pcb with print layout 6.0 and sent the gerber files to a pcb manufacturer in China and everything has been under control. When you had built your 2 AXLs, did you match the output mosfets? I have noticed that If you don't match them, there is a problem in adjusting the quiescent current around 1A for both channels for class A operation! In my case the 1st channel could reach at 980mA and the 2nd just 800mA maximum. That was a big problem for the sonic performance of the amplifier. So I have adjusted 800mA for both. My second AXL construction (2 months ago) is running perfectly because I purchased 2 pairs of matching mosfets( 2SJ49 - 2SK134) and the quiescent current was easily adjusted at 1A. Its sonic performance is increadible. Its power supply is consisted of 40.000uF + 40.000uF and there is a voltage stabilization circuit with 2N6287 and 2N6284 darlington transistors for +-30volts - 10Amperes. The measured ripple is less than 100microVolts.I can post some images if you want.Thanks for your reply.</p></div> <div data-bbox="459 1028 635 1061">Report Post</div> |
| <div data-bbox="253 1196 288 1254"><b>J</b></div> <div data-bbox="212 1305 331 1339"><b>johnego</b></div> <div data-bbox="173 1355 276 1379">Member</div> <div data-bbox="156 1391 288 1415">Joined 2017</div> | <div data-bbox="459 1173 746 1205">2021-04-10 6:01 pm</div> <div data-bbox="1278 1173 1453 1205">  #29</div> <div data-bbox="496 1270 1366 1305"><b><u>metanastis said:</u></b> (<i>URL: /community/goto/post?id=6610845</i>)</div> <div data-bbox="496 1337 987 1370">Its sonic performance is increadible.</div> <div data-bbox="459 1462 1453 1541"><p>How do you compare this AXL with other Latfet amplifiers? (Hitachi, JLH)</p></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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| <p><b>metanastis</b><br/>Member<br/>Joined 2009</p>           | <p>2021-04-10 7:27 pm <span>🔗 📌 #30</span></p> <p>I have constructed just 2 AXL amps and this one(Lang 20W class A) will be my 3rd. All of these 3 aps use the same hitachi mosfets 2SJ49-2Sk134. I haven't tried any other brand of mosfets. These hitachi mosfets have got a great reputation through the time and I trust them.Herr Flick had made a modification to the old AXL circuit using lateral mosfet of Exicon. He said that their sonic performance was great, eventhough it operates in class AB.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p><b>J</b><br/><b>johnego</b><br/>Member<br/>Joined 2017</p> | <p>2021-04-11 3:55 am <span>🔗 📌 #31</span></p> <div data-bbox="464 712 1453 1099"><p><b><u>metanastis said:</u></b> (<a href="/community/goto/post?id=6611462">URL: /community/goto/post?id=6611462</a>)</p><p>I have constructed just 2 AXL amps and this one(Lang 20W class A) will be my 3rd. All of these 3 aps use the same hitachi mosfets 2SJ49-2Sk134. I haven't tried any other brand of mosfets. These hitachi mosfets have got a great reputation through the time and I trust them.Herr Flick had made a modification to the old AXL circuit using lateral mosfet of Exicon. He said that their sonic performance was great, eventhough it operates in class AB.</p></div> <p>Thanks. Yes I heard the newer mosfet is better sounding. And AXL had a good reputation too in the past, just like the Crescendo. But today we have more options that I found almost no one favor the amps anymore.</p> <p>Actually, in the past many people didn't like the sound of mosfet amps when side by side compared to BJT amps. It's like comparing TDA7294 with LM1875 you will see that the majority will lean towards the latest.</p> <p>I have always wanted to build this AXL just to know how it would sound like but I knew it would be a waste of time. I guess the positive feedback will have a positive effect on damping hence better bass (e.g. than Crescendo), but I don't know in reality. What I want to know is the quality in other area other than bass.</p> <p>BTW, I will google the Lang 20W.</p> |

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| <p><b>metanastis</b><br/>Member<br/>Joined 2009</p>           | <p>2021-04-11 8:51 am <span>↩</span> <span>🔖</span> #32</p> <p>A class A amplifier has the best sonic performance from any other transistor ampifier thanks to its principals of operation. There are surely better class A amplifiers than AXL. When I built my 1st AXL back in 1988, I was astonished by its performance. I had compared that to my friends' expensive BJT ones and mine was far better. Give it a try and it won't let you down! I mean it !!</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b>J</b><br/><b>johnego</b><br/>Member<br/>Joined 2017</p> | <p>2021-04-11 5:55 pm <span>↩</span> <span>🔖</span> #33</p> <div data-bbox="462 665 1453 1016"><p><b>metanastis said:</b> (<i>URL: /community/goto/post?id=6612083</i>)</p><p>A class A amplifier has the best sonic performance from any other transistor ampifier thanks to its principals of operation. There are surely better class A amplifiers than AXL. When I built my 1st AXL back in 1988, I was astonished by its performance. I had compared that to my friends' expensive BJT ones and mine was far better. Give it a try and it won't let you down! I mean it !!</p></div> <p>Thx for the encouragement hehe. Wait, is AXL class-A? Or did you mean it can be biased more into class-A?</p> <p>Have you heard the VSSA? Many people like the sound of VSSA but my bipolar amp is much better. But I know that better class-B amp can be made with latfet so I kind of desperate searching for basic latfet amp that I can improve to better my bipolar one.</p> |
| <p><b>metanastis</b><br/>Member<br/>Joined 2009</p>           | <p>2021-04-11 10:49 pm <span>↩</span> <span>🔖</span> #34</p> <p>By adjusting the quiescent current potentiometer you can adjust the class of the amplifier. At 350mA you will get a class AB amplifier. At 1A you get pure class A. You may choose 750-800mA for a mixed class A till 10Watts and class AB for the rest. It's up to you.</p> <p>If you need the full article from elektor I can send you a pdf. To my opinion, the AXL is a much better amplifier than any BJT amplifier. If you can afford of an about 500 euros expense you mut give it a try!!!</p>                                                                                                                                                                                                                                                                                                                                                                                                         |

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| <div data-bbox="252 174 288 230">J</div> <div data-bbox="212 282 331 315">johnego</div> <div data-bbox="159 331 288 392">Member<br/>Joined 2017</div>        | <div data-bbox="461 152 743 185">2021-04-12 1:26 am</div> <div data-bbox="1278 152 1453 185">  #35</div> <div data-bbox="496 248 1367 284"><b>metanastis said:</b> (<i>URL: /community/goto/post?id=6613231</i>)</div> <div data-bbox="496 313 1414 468">If you need the full article from elektor I can send you a pdf. To my opinion, the AXL is a much better amplifier than any BJT amplifier. If you can afford of an about 500 euros expense you must give it a try!!!</div> <div data-bbox="461 560 1434 595">Ok thank you, I will give it a try. And yes I want the article, thanks...</div> <div data-bbox="461 647 1418 779">I wont need 500 euros as I have all the parts. For testing I will use TO3P. When it has to be TO3, creating the holes on the heatsink and cutting the aluminium are things I don't really like</div> <div data-bbox="461 833 1398 869">I will compare it side by side with the NRDS by J.M. Plantefeve...</div> |
| <div data-bbox="189 1111 354 1142">metanastis</div> <div data-bbox="159 1160 292 1220">Member<br/>Joined 2009</div>                                          | <div data-bbox="461 978 758 1012">2021-04-12 11:42 am</div> <div data-bbox="1278 978 1453 1012">  #36</div> <div data-bbox="461 1052 1445 1178">The most expensive parts are the heat sinks, chassis and mosfets. The rest are really cheap. The pdf is too large. I will try to send it by pm.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <div data-bbox="252 1332 300 1388">K</div> <div data-bbox="201 1440 344 1473">kannan_s</div> <div data-bbox="159 1489 292 1550">Member<br/>Joined 2009</div> | <div data-bbox="461 1308 758 1341">2021-04-12 12:21 pm</div> <div data-bbox="1278 1308 1453 1341">  #37</div> <div data-bbox="461 1382 1305 1550">Hi Metanastis<br/>can you also please send me the copy of the PDF. Thanks<br/>thanks<br/>kannan</div> <div data-bbox="461 1626 662 1657">From Austin TX</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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| <p><b>metanastis</b></p> <p>Member</p> <p>Joined 2009</p>              | <p>2021-04-12 2:05 pm <span>🔗 📌 #38</span></p> <p>That's the full article from the elektor english version for everyone it may concern.</p> <div data-bbox="461 342 1453 891"><p><b>Attachments</b></p><div data-bbox="523 488 639 629"></div><p><u>(URL: /community/attachments/pages-from-elektor-1985-03-1-pdf.941635/)</u></p><p>Pages from Elektor-1985-03-1.pdf</p><p>592.8 KB · Views: 459</p></div>           |
| <p><b>J</b></p> <p><b>johnego</b></p> <p>Member</p> <p>Joined 2017</p> | <p>2021-04-13 5:29 am <span>🔗 📌 #39</span></p> <div data-bbox="461 1066 1453 1256"><p><b><u>metanastis said:</u></b> (URL: /community/goto/post?id=6613955)</p><p>That's the full article from the elektor english version for everyone it may concern.</p></div> <p>Thank you, metanastis! So the AXL was intended for class-A with 350mA as the compromise between heat and quality. This is interesting because increasing bias in Crescendo didn't make it sound better (at least to my ears).</p> |
| <p><b>metanastis</b></p> <p>Member</p> <p>Joined 2009</p>              | <p>2021-04-13 7:59 am <span>🔗 📌 #40</span></p> <p>Believe me, for 350mA to 1A the sonic difference is audible !!</p>                                                                                                                                                                                                                                                                                                                                                                                   |

J

**johnego**Member  
Joined 2017

2021-04-21 12:21 am

  #41

Hi metanastis and HerrFlick,

My progress so far and I have a question. How hot is the driver transistor? The PCB is quite small and I had to find suitable heatsink for that. Will the heatsink in the picture be sufficient? And if I put a rail capacitor in front of it (touching the heatsink) will it be a problem? Otherwise I need to find/make a heatsink that will extend off board to give room for the capacitor.

**Attachments**

([URL: /community/attachments/axl\\_photo1-jpg.944462/](/community/attachments/axl_photo1-jpg.944462/))

AXL\_photo1.jpg

146.2 KB · Views: 326

H

**HerrFlick**Member  
Joined 2017

2021-04-21 9:41 am

  #42

Hello JOHNEGO,

It all depends of the quiescent current you are using. I have a infrared temperature probe so I measured the temperature of my amplifier using a quiescent current of 0,2 A and all four driver transistors are about 25 °C. The heatsink is 12 °C/W.

|                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <div data-bbox="252 174 288 230">J</div> <div data-bbox="212 282 331 315">johnego</div> <div data-bbox="172 331 274 356">Member</div> <div data-bbox="158 369 288 392">Joined 2017</div>           | <div data-bbox="461 152 751 181">2021-04-21 11:21 am</div> <div data-bbox="1278 152 1453 181">  #43</div> <div data-bbox="496 248 1339 284"><b>HerrFlick said:</b> (URL: /community/goto/post?id=6625819)</div> <div data-bbox="496 315 1393 508">Hello JOHNEGO,<br/>It all depends of the quiescent current you are using. I have a infrared temperature probe so I measured the temperature of my amplifier using a quiescent current of 0,2 A and all four driver transistors are about 25 °C. The heatsink is 12 °C/W.</div> <div data-bbox="461 600 1447 768">Hello HerrFlick, thanks. 200mA is below the optimum stated by the document which is 350mA and metanastis uses 0.6-1A which he said is much better; so there will be big possibility that I will try that class A approach...</div> <div data-bbox="461 824 1422 945">May be I will start with 200mA first and then learn from that. Or, I can use bigger heatsink but put it (with the transistor) hanging on the bottom site of the PCB</div> |
| <div data-bbox="248 1081 296 1137">H</div> <div data-bbox="207 1189 341 1220">HerrFlick</div> <div data-bbox="172 1238 274 1263">Member</div> <div data-bbox="158 1276 288 1299">Joined 2017</div> | <div data-bbox="461 1059 758 1088">2021-04-21 11:39 am</div> <div data-bbox="1278 1059 1453 1088">  #44</div> <div data-bbox="461 1133 1058 1167">What kind of power mosfets do you use?</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <div data-bbox="252 1411 288 1467">J</div> <div data-bbox="212 1518 331 1552">johnego</div> <div data-bbox="172 1568 274 1592">Member</div> <div data-bbox="158 1606 288 1628">Joined 2017</div>   | <div data-bbox="461 1388 740 1417">2021-04-21 2:11 pm</div> <div data-bbox="1278 1388 1453 1417">  #45</div> <div data-bbox="496 1485 1339 1520"><b>HerrFlick said:</b> (URL: /community/goto/post?id=6625909)</div> <div data-bbox="496 1552 1061 1583">What kind of power mosfets do you use?</div> <div data-bbox="461 1677 1441 1798">Many. Mostly K135/J50. But the ones on the picture are BIPOLARS. Just there for part placement planning. Latfets are sensitive to electrostatics so I used bipolars for this purpose.</div> <div data-bbox="461 1854 1433 1977">BTW the Rout are too close to heatsink. I may lift its position to avoid heating (if this is an issue). Currently it's 0.15 Ohm but later I may reduce the value as I have down to 0.005 Ohm.</div>                                                                                                                                                                                                                                 |

H

**HerrFlick**Member  
Joined 2017

2021-04-21 3:11 pm

  #46

Ok, I'am using ECF10N20/ECF10P20 in class AB, the application notes from Profusion tells me to use 0,1-02 A. There is a very interesting application note from Texas Instruments AN-1645 - Driving a MOSFET Output stage- I can recommend that to you. 27 pages of info about using 2SK1058/2SJ162, BUZ901/BUZ906, 2SK1530/2SJ201 and IRFP240/IRFP9240. How to calculate a Snubber network, Bias etc.

J

**johnego**Member  
Joined 2017

2021-04-21 3:34 pm

  #47**HerrFlick said:** (URL: /community/goto/post?id=6626097)

Ok, I'am using ECF10N20/ECF10P20 in class AB, the application notes from Profusion tells me to use 0,1-02 A. There is a very interesting application note from Texas Instruments AN-1645 - Driving a MOSFET Output stage- I can recommend that to you. 27 pages of info about using 2SK1058/2SJ162, BUZ901/BUZ906, 2SK1530/2SJ201 and IRFP240/IRFP9240. How to calculate a Snubber network, Bias etc.

Yes, I think you have put a link to that document here somewhere. Latfets have zero temp coef. around 100ma+ so without heatsinking and class A consideration 100mA-200mA is about right. But it doesn't mean that it is the best approach. But it is strange that higher bias with latfets has usually little benefit if any. My preferred device for class A is actually HEXFETs where in class A their capacitance is not a big issue but there is clear benefit from their higher transconductance. May be an array of latfets is as good but that will be expensive.

None of these latfets pairs you mentioned are balanced between its channels. So balancing seems logical. But I have never observed improvement in sound quality with balancing the junction capacitances so I don't go the trouble of adding extra resistances or capacitances (the snubbers).

Last edited: 2021-04-21 3:40 pm

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| <b>metanastis</b><br>Member<br>Joined 2009                                                                                   | 2021-04-22 1:50 pm   #48<br><br>Try to adjust the quiescent current more than 750mA and listen to the different sound of that great amplifier. I have and I can assure you that there is a great difference between class AB from class A. If you use 2SK1233/34/135 & 2SJ48/49/50 you must match them. Otherwise you will not be able to increase the current more that 700mA t both channels.                                                                                                                                                                                       |
| <br><b>johnego</b><br>Member<br>Joined 2017 | 2021-04-24 5:34 am   #49<br><div><b>metanastis said:</b> (<a href="/community/goto/post?id=6627419">URL: /community/goto/post?id=6627419</a>)<br/>Try to adjust the quiescent current more than 750mA and listen to the different sound of that great amplifier. I have and I can assure you that there is a great difference between class AB from class A.</div><br>Unlike Pass Labs' amplifiers, IME amps like this requires preamp to sound good. Otherwise it will sound rather "dull". But I don't usually use preamps. I have only DCB1 (buffer) here. What preamp do you use? |
| <b>metanastis</b><br>Member<br>Joined 2009                                                                                   | 2021-04-24 5:56 am   #50<br><br>I use a clone of the great Marantz 7C revision 7 tube line preamplifier made by myself and it sounds great!                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| <p><b>J</b></p> <p><b>johnego</b></p> <p>Member<br/>Joined 2017</p> | <p>2021-04-24 1:31 pm <span>↩</span> <span>🔖</span> #51</p> <div data-bbox="464 224 1453 412"><p><b><u>metanastis said:</u></b> (URL: /community/goto/post?id=6629755)</p><p>I use a clone of the great Marantz 7C revision 7 tube line preamplifier made by myself and it sounds great!</p></div> <p>I built the Marantz 20 years ago. I didn't like the sound at all. Still have it somewhere stuffed with fancy riken gold and black gates. There were many single stage mu-follower circuits on the web that sound better.</p>                                                                                                                                                    |
| <p><b>metanastis</b></p> <p>Member<br/>Joined 2009</p>              | <p>2021-04-25 3:41 pm <span>↩</span> <span>🔖</span> #52</p> <p>Mine has just an outstanding performance using NOS american tubes .It uses1 and half tubes per channel. I have copied many tube preamps like Audio Note, Cary, Audio Reseach etc. Marantz 7c is my best so far!</p> <p>Last edited: 2021-04-25 3:44 pm</p>                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>J</b></p> <p><b>johnego</b></p> <p>Member<br/>Joined 2017</p> | <p>2021-04-26 7:35 am <span>↩</span> <span>🔖</span> #53</p> <div data-bbox="464 1256 1453 1529"><p><b><u>metanastis said:</u></b> (URL: /community/goto/post?id=6631238)</p><p>Mine has just an outstanding performance using NOS american tubes .It uses1 and half tubes per channel. I have copied many tube preamps like Audio Note, Cary, Audio Reseach etc. Marantz 7c is my best so far!</p></div> <p>Yes, many liked the 7C but I didn't. I guess because my speaker was not suitable with tube stuffs. Or I just don't like any apparent extra noise added by any extra stages... Or I just don't like distortion... (but I have one tube stage that I like, after a DAC)</p> |
| <p><b>J</b></p> <p><b>johnego</b></p>                               | <p>2021-05-09 12:24 pm <span>↩</span> <span>🔖</span> #54</p> <p><b>AXL Amp Review</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Member

Joined 2017

AXL is an old amp, almost as old as Crescendo. But it took me time to try the AXL because many people said that it is comparable to the (mini) Crescendo. But now I know that it is to my taste better than the Crescendo, and especially when biased high, it is absolutely better.

AXL is better than I expected, but cannot compete with my other amps.

So I modified the AXL using all small signal transistor to copy the idea from NRDS amp by J.M Plantefevé who "modified" the Crescendo also using small signal transistors. The feedback return point is changed as used by common amps and there are many other modifications.

I attached the FFT of the AXL, my AXL mod and NRDS. I believe that most of the amplifiers sound quality can be seen from simulation. And from simulation, most of the quality can be seen from FFT.

The NRDS distortion is higher but it is in the style of the Hiraga super class-A that even though high distortion but many people like it and consider it as one of the best sounding amplifier. I have another amp that is better than the NRDS and have the same distortion spectrum but like the Hiraga, the bias is twice as high (and when I reduce the bias to 1.1A it is not as good as the NRDS).

My AXL mod doesn't look special in the FFT (even though it is better than the original) but it has something else that cannot be seen from the FFT.

Note: No simulation file will be posted.

### Attachments

[\(URL: /community/attachments/axl-png.950180/\)](https://www.diyaudio.com/community/attachments/axl-png.950180/)

AXL.png

49.3 KB · Views: 244

**W****wahab**

Member

Joined 2009

2021-06-11 11:27 pm

  #55

Not that i like to criticise but i simulated extensively Elektor s Crescendo and most of its later iterations including the AXL and frankly that s quite poorly implemented designs.

The crescendo particularly, where they explain that they use some kind of shunt compensation by the virtue of the input capacitance of the lateral fets loading the "VAS", but it simply cant work due to the high standing current of the driving stage, hence an hasardous stability...

Also the AXL is an inverting amp and contrary to what is generally thought this has an importance and require to invert the speakers terminals, that is, to connect the plus of the speakers to the the amp s ground.

**sercan85**

Member

Joined 2016

Thursday at 7:20 AM

New   #56

hello

I would like to build this amp with ECX10N20S and 10P20S. ECF10n20 and 10p20 are End of life and little expensive.

[metanastis](#)

|                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <div data-bbox="188 286 354 315">metanastis</div> <div data-bbox="173 336 276 360">Member</div> <div data-bbox="158 371 292 396">Joined 2009</div>       | <div data-bbox="461 154 758 185">Yesterday at 4:24 PM</div> <div data-bbox="1182 159 1453 185">New   #57</div> <div data-bbox="496 250 1299 286"><b>wahab said:</b> (URL: /community/goto/post?id=6688071)</div> <div data-bbox="496 318 1380 430">Not that i like to criticise but i simulated extensively Elektor s Crescendo and most of its later iterations including the AXL and frankly that s quite poorly implemented designs.</div> <div data-bbox="496 479 1417 631">The crescendo particularly, where they explain that they use some kind of shunt compensation by the virtue of the input capacitance of the lateral fets loading the "VAS", but it simply cant work due to the high standing current of the driving stage, hence an hasardous</div> <div data-bbox="461 676 1431 976">Hello,<br/>If you read the old Elektor article, there is no reference in any part of it that the output is inverted.<br/>When I built that amp, I used an audio signal generator and the signal at the output was at the same phase as the input.<br/>Have you tested that amp and have seen the inverted signal at the output?</div> |
| <div data-bbox="188 1223 354 1252">metanastis</div> <div data-bbox="173 1270 276 1294">Member</div> <div data-bbox="158 1305 292 1330">Joined 2009</div> | <div data-bbox="461 1090 758 1122">Yesterday at 4:37 PM</div> <div data-bbox="1182 1095 1453 1122">New   #58</div> <div data-bbox="496 1187 1342 1223"><b>sercan85 said:</b> (URL: /community/goto/post?id=7629008)</div> <div data-bbox="496 1252 564 1281">hello</div> <div data-bbox="496 1332 1313 1406">I would like to build this amp with ECX10N20S and 10P20S.<br/>ECF10n20 and 10p20 are End of life and little expensive.</div> <div data-bbox="461 1453 1452 1711">The AXL has been a great amp in its era. That was my 1st amp I built when I was a student and it has worked perfectly in all of my dj activities( I had to lower the QC to get more power). I have read a lot of good comments for the Exicon mosfets and as HerrFlick has written, they are better than the old 2SK and 2SJ ones. It's up to you.</div>                                                                                                                                                                                                                                                                                                 |

**wahab**

Member

Joined 2009

Yesterday at 6:56 PM

New



#59

**metanastis said:** ([URL: /community/goto/post?id=7630261](/community/goto/post?id=7630261))

Hello,

If you read the old Elektor article, there is no reference in any part of it that the output is inverted.

When I built that amp, I used an audio signal generator and the signal at the output was at the same phase as the input.

Have you tested that amp and have seen the inverted signal at the output?

Here the schematic of the AXL, that s a phase inverting amp, the input signal is sent on the inverting input and the non inverting input is grounded AC and DC wise, to retain the real phase in respect of the audio signal the speakers connection should be inverted, guess that you confused it with the Mini Crescendo wich is not an inverting phase amp.

### Attachments

[\(URL: /community/attachments/axl-v2-schema-jpg.1286491/\)](/community/attachments/axl-v2-schema-jpg.1286491/)

AXL-v2-schema.jpg

294.1 KB · Views: 11

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